Credible Government Policies with Reputations

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Introduction

- A *government policy* is a sequence of government actions in every history
- A government policy is *credible* if it is in the government's interest to implement actions prescribed by it, conditional on every possible history
- This chapter: a government policy can be rendered credible by a form of **history dependence** (i.e. reputation): future outcomes depend on past actions

Outline

1. A one-period economy
   - A "game" interpretation
   - Ramsey and Nash outcomes
2. An infinitely-repeated economy
   - Subgame Perfect Equilibrium (SPE)
   - APS and recursive strategies
   - Compute the SPE set
3. Example: a taxation problem
4. Conclusion and extension
   - Two sources of history dependence
   - Alternative way to get credibility

A one-period economy

Environment

- A continuum of identical households with utility $u(\xi, x, y)$
  - Households choose $\xi \in X$, given x and y
  - Aggregate action: $x = \int \xi(i)\, di \in X$
- A government chooses an action $y \in Y$

Definition 1 (Competitive equilibrium given y)

A *competitive equilibrium given* y is $x = h(y) \in X$ such that:

1. Given anticipated x and government action y, households solve $\max_{\xi \in X} u(\xi, x, y)$.
2. $x = \arg\max_{\xi \in X} u(\xi, x, y)$.

Let $C = \{(x, y) \mid x = h(y)\}$ be the set of competitive equilibria.

- Assume X, Y are compact, and $u(.)$ is continuous. Then C is compact.

Two timing protocols

- **Timing protocol 1**
  1. The government chooses $y \in Y$
  2. Given y, the private sector forms a competitive equilibrium $(h(y), y)$

Anticipating the private sector's response, the government solves a *Ramsey problem*: $\max_{y \in Y} u(h(y), h(y), y)$.

- **Timing protocol 2**
  1. Given some belief y, the private sector forms a competitive equilibrium (x, y) with $x = h(y)$
  2. Given x, the government chooses $\eta \in Y$ to maximize $u(x, x, \eta)$

If the government's optimal action is $\eta^* = y$, we can **interpret** the outcome as a *Nash equilibrium*.

"Nash" equilibrium

Definition 2 (Nash equilibrium)

A *Nash equilibrium* is (x^N, y^N) that satisfies:

1. $(x^N, y^N) \in C$.
2. Given x^N, $y^N = \arg\max_{\eta \in Y} u(x^N, x^N, \eta) \equiv H(x^N)$.

- Recall from game theory in a simultaneous-move game:
  - Each player **simultaneously** chooses best response given her belief about the other players' actions.
  - In a Nash equilibrium, beliefs match actual actions.

- Here, a different timing protocol:
  - Households move first, but **have no first-mover advantage** due to CE; hence we can interpret $h(y)$ as their "best reponse" given belief y.
  - We look for equilibrium where it's in the government's interests to confirm private sector beliefs, i.e., $H(x)$ is its "best response" given observed x.

Example: a taxation problem

- Environment:
  - Endowment: each household has one unit of time
  - Technology: output is linear in time with slope 1
  - Given (τ, g), household maximizes:

$$U(\ell, c, g) = \ell + \log(\alpha + c) + \log(\alpha + g), \qquad \alpha \in (0, \tfrac{1}{2})$$

    subject to $c = (1-\tau)(1-\ell)$, with $\ell \in [0, 1]$
  - Government budget: $g = \tau(1 - \bar{\ell})$, where $\bar{\ell}$ is average leisure and $\tau \in [0, 1]$

- Optimal leisure decision:

$$\ell(\tau) = \begin{cases} \dfrac{\alpha}{1-\tau}, & 0 \le \tau \le 1 - \alpha, \\ 1, & 1 - \alpha < \tau \le 1. \end{cases}$$

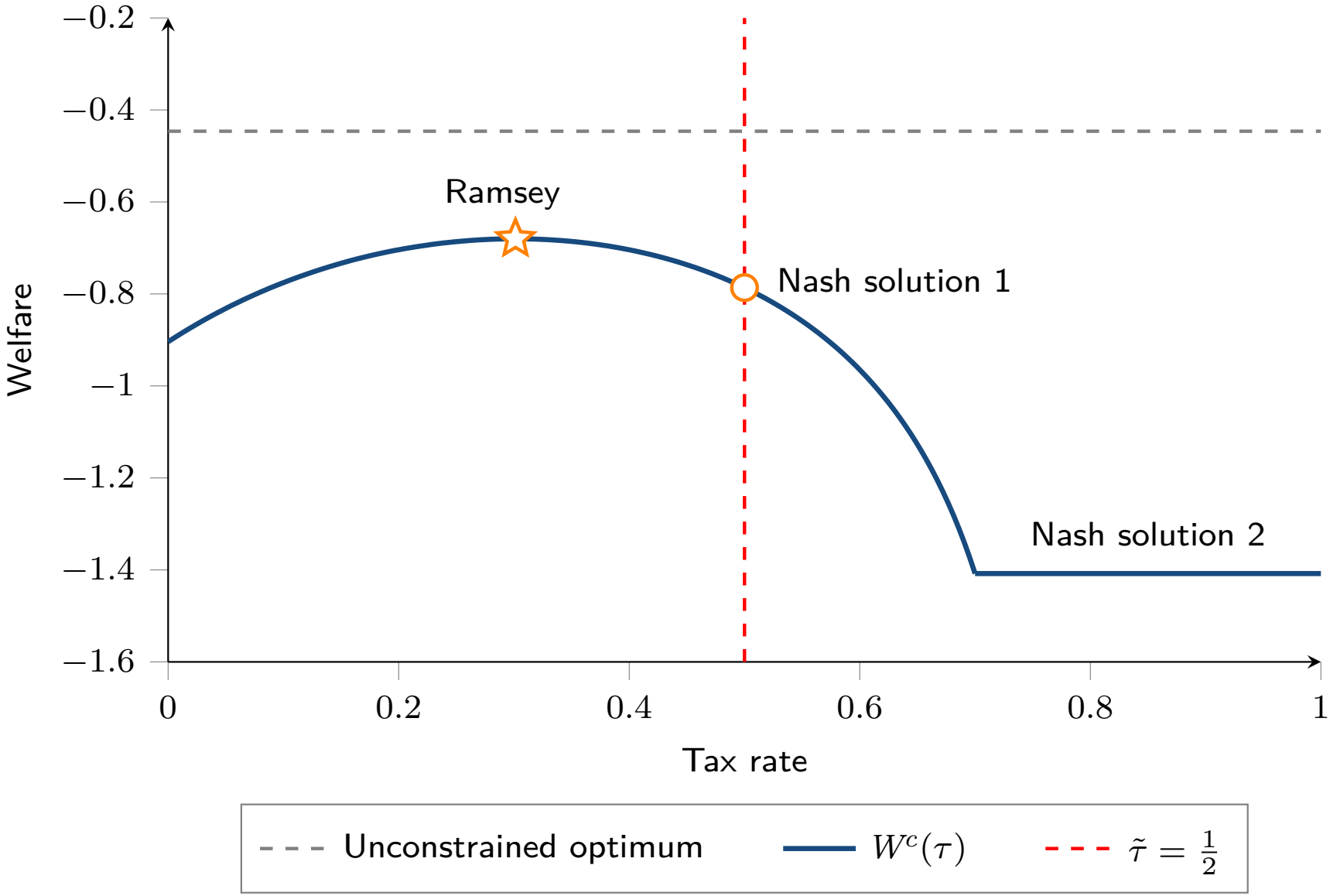
Taxation: planner, Ramsey, and Nash

- Planner's problem:
  - $\max_{\ell,c,g} U(\ell, c, g)$ subject to $\bar{c} + g = 1 - \bar{\ell}$.
  - Solution: $c^P = g^P = \frac{1}{2}$, $\ell^P = 0$.
- Ramsey problem:
  - $\max_\tau W^c(\tau) \equiv \ell(\tau) + \log\{\alpha + (1-\tau)[1-\ell(\tau)]\} + \log\{\alpha + \tau[1-\ell(\tau)]\}$
  - Solution: $\tau^R \in (0, \frac{1}{2})$ solved by

$$[1 - \ell(\tau^R)] \left[\frac{1}{\alpha + g^R} - \frac{1}{\alpha + c^R} \right] = \frac{\tau^R \ell'(\tau^R)}{\alpha + g^R}.$$

  - **Time inconsistency**: after households choose $\ell = \ell^R$, optimal to reset $\tilde{\tau} = \frac{1}{2}$.
- Nash equilibrium:
  - $\ell^N = \ell(\tau^N)$, $\tau^N \in \arg\max_\tau \{\ell^N + \log[\alpha + (1-\tau)(1-\ell^N)] + \log[\alpha + \tau(1-\ell^N)]\}$.
  - Solution 1: $\ell^N < 1$, $\tau^N = \frac{1}{2}$, $c^N = g^N = \frac{1}{2} - \alpha$.
  - Solution 2: $\ell^N = 1$, $\tau^N \in [1-\alpha, 1]$, $c^N = g^N = 0$. Later: set $\tau \in [0, 1 - \alpha - \epsilon]$.

Taxation: $\alpha = 0.3$



An infinitely repeated economy

Strategy

Repeat the one-period economy forever, with $x^t = \{x_s\}_{s=1}^t$, $y^t = \{y_s\}_{s=1}^t$.

Definition 3 (Strategy)

A *pure strategy* is a sequence of functions $\{\sigma_t^h\}_{t=1}^\infty$, $\{\sigma_t^g\}_{t=1}^\infty$ mapping the history (x^{t-1}, y^{t-1}) into an action at t.

- For the aggregate of households, $\sigma^h = \{\sigma_t^h\}_{t=1}^\infty$, with

$$\sigma_1^h \in X, \qquad \sigma_t^h : X^{t-1} \times Y^{t-1} \to X \quad \text{for each } t \geq 2.$$

- For the government, $\sigma^g = \{\sigma_t^g\}_{t=1}^\infty$, with

$$\sigma_1^g \in Y, \qquad \sigma_t^g : X^{t-1} \times Y^{t-1} \to Y \quad \text{for each } t \geq 2.$$

Definition 4 (Strategy profile)

A *strategy profile* is $\sigma = (\sigma^h, \sigma^g)$, with $\sigma_t = (\sigma_t^h, \sigma_t^g)$.

Continuation strategy

Definition 5 (Continuation strategy)

A *continuation strategy* is the strategy profile $\sigma|_{(x^t,y^t)}$ obtained by fixing history (x^t, y^t) in σ and treating date $t+1$ as period 1:

$$\left(\sigma|_{(x^t,y^t)}\right)_1 = \sigma_{t+1}(x^t, y^t).$$

For every $s \geq 1$ and every subsequent history $(\nu^s, \eta^s) \in X^s \times Y^s$,

$$\begin{aligned}&\left(\sigma|_{(x^t,y^t)}\right)_{s+1}(\nu^s, \eta^s)\\ &\quad = \sigma_{t+s+1}(x_1, \ldots, x_t, \nu_1, \ldots, \nu_s;\ y_1, \ldots, y_t, \eta_1, \ldots, \eta_s).\end{aligned}$$

A strategy profile implies a history and a value

- A strategy profile σ recursively generates a history:

$$[x(\sigma)_1, y(\sigma)_1] = (\sigma_1^h, \sigma_1^g),$$
$$[x(\sigma)_t, y(\sigma)_t] = \sigma_t[x(\sigma)^{t-1}, y(\sigma)^{t-1}], \quad t \geq 2.$$

- The value to the government associated with σ is

$$V_g(\sigma) = \frac{1-\delta}{\delta} \sum_{t=1}^{\infty} \delta^t r[x(\sigma)_t, y(\sigma)_t].$$

  - $r(x_t, y_t) \equiv u(x_t, x_t, y_t)$ is the government's one-period payoff.
  - $0 < \delta < 1$ is the discount factor.

- Recursive representation:

$$V_g(\sigma) = (1-\delta) r(x_1, y_1) + \delta V_g\big(\sigma|_{(x_1, y_1)}\big).$$

Subgame perfect equilibrium (SPE)

Definition 6 (Subgame perfect equilibrium)

A strategy profile $\sigma = (\sigma^h, \sigma^g)$ is a *subgame perfect equilibrium* (SPE) of the infinitely repeated economy if for each $t \geq 1$ and each history $(x^{t-1}, y^{t-1}) \in X^{t-1} \times Y^{t-1}$

1. The private sector outcome $x_t = \sigma_t^h(x^{t-1}, y^{t-1})$ is consistent with competitive equilibrium when $y_t = \sigma_t^g(x^{t-1}, y^{t-1})$;
2. For each possible government action $\eta \in Y$

$$(1-\delta)r(x_t, y_t) + \delta V_g\big(\sigma|_{(x^t, y^t)}\big) \geq (1-\delta)r(x_t, \eta) + \delta V_g\big(\sigma|_{(x^t;\, y^{t-1}, \eta)}\big).$$

Recursive characterization of SPE

Proposition 1 (Continuation strategy profile is SPE)

Assume that σ is an SPE. Then for all $(\nu, \eta) \in X \times Y$, $\sigma|_{(\nu,\eta)}$ is an SPE.

Lemma 1 (Recursive characterization)

Consider a strategy profile σ, and let the associated first-period outcome be given by $x = \sigma_1^h$, $y = \sigma_1^g$. The profile σ is an SPE if and only if

1. *for each $(\nu, \eta) \in X \times Y$, $\sigma|_{(\nu,\eta)}$ is an SPE;*
2. *(x, y) is a competitive equilibrium;*
3. *$\forall\, \eta \in Y$,*

$$(1-\delta) r(x,y) + \delta\, V_g\big(\sigma|_{(x,y)}\big) \geq (1-\delta) r(x,\eta) + \delta\, V_g\big(\sigma|_{(x,\eta)}\big).$$

Taxation: repeated Nash and Nash trigger

Set $\alpha = 0.3$, $\varepsilon = 0.1$, $\delta = 0.20$. The unique NE is $(\ell^N, \tau^N) = (0.6, \frac{1}{2})$.

- Infinite repetition of Nash
  - $\sigma_t^N = (\ell^N, \tau^N)$ at every date and every history.
  - The continuation value is always $v^N = W^c(\tau^N) = -0.7863$.

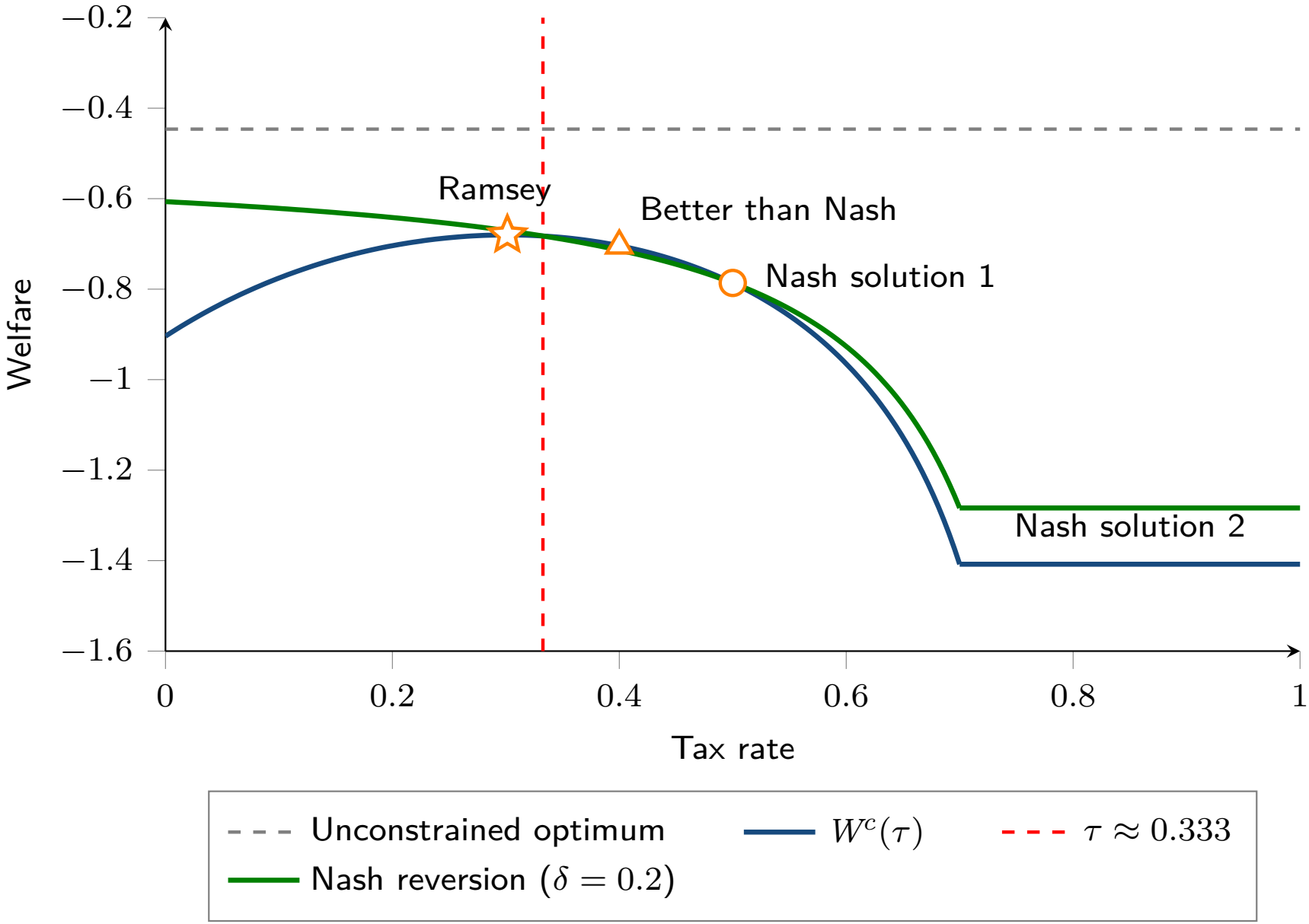
- A trigger strategy:
  - **Repeat** $(\ell(0.4), 0.4)$ **if all past outcomes equal this pair**; otherwise repeat Nash forever, which is SPE.
  - Incentive constraint holds:

$$\underbrace{W^c(0.4)}_{-0.7040} \geq (1-\delta) \underbrace{\max_{0 \leq \eta \leq 0.6} r[\ell(0.4), \eta]}_{-0.6957} + \delta v^N = -0.7138.$$

  - However, fails to sustain Ramsey:

$$\underbrace{W^c(\tau^R)}_{-0.6801} < (1-\delta) \underbrace{\max_{0 \leq \eta \leq 0.6} r[\ell(\tau^R), \eta]}_{-0.6418} + \delta v^N = -0.6707.$$

Taxation: Nash trigger



Continuation strategies $\Longleftrightarrow$ values

- The set of SPE values:

$$V = \{ V_g(\sigma) \mid \sigma \text{ is an SPE} \}.$$

Lemma 2 ($v(.)$ characterization)

A competitive equilibrium $(x, y) \in C$ is supported by an SPE, i.e. $x = \sigma_1^h$ and $y = \sigma_1^g$, if and only if there exists a function $v(\cdot) : Y \to V$ such that

$$(1-\delta) r(x, y) + \delta v(y) \geq (1-\delta) r(x, \eta) + \delta v(\eta), \qquad \forall \eta \in Y.$$

Proof.

- *Necessity*: SPE continuation strategies produce $(v(y), v(\eta)) \in V$.
- *Sufficiency*: Given such $(v(y), v(\eta)) \in V$, use SPEs attaining these values as continuation strategies; apply Lemma 1.

APS machinery

Can we find (or characterize) all SPEs?

Abreu, Pearce, and Stacchetti (1990) finds a recursive procedure that can recover all SPE values.

- Useful to us because it tells us whether Ramsey can be sustained by *any* credible punishment.

APS machinery

Definition 7 (W-admissible CE-value pair)

Let $W \subset \mathbb{R}$. A triple $(x, y, w(\cdot))$ is said to be *admissible with respect to* W if $(x, y) \in C$ and $w(\cdot) : Y \to W$ is a function satisfying

$$(1-\delta)r(x,y) + \delta w(y) \geq (1-\delta)r(x,\eta) + \delta w(\eta), \qquad \forall \eta \in Y.$$

Definition 8 (Operator B)

For each set $W \subset \mathbb{R}$, let $B(W)$ be the set of possible values

$$v = (1-\delta)r(x,y) + \delta w(y)$$

for which **there exists** some admissible triples $(x, y, w(\cdot))$ with respect to W.

APS machinery

Definition 9 (Self-generating set)

The set W is said to be *self-generating* if $W \subseteq B(W)$.

Equivalently, for every $v \in W$, there exists $(x, y, w(\cdot))$ satisfying

$$v = (1-\delta)r(x,y) + \delta w(y) \geq (1-\delta)r(x,\eta) + \delta w(\eta), \qquad \forall \eta \in Y.$$

- Properties of $B(\cdot)$:
  - Monotonicity: If $W \subseteq W' \subseteq \mathbb{R}$, then $B(W) \subseteq B(W')$.
  - $B(W)$ is compact if W is compact.

APS machinery

Theorem 1 (A self-generating set is a subset of V)

If $W \subset \mathbb{R}$ is bounded and self-generating, then $B(W) \subseteq V$.

In words, a set of promises becomes **credible** when every promise in it can be supported by some promises from the same set.

Proof.

- Pick $v \in B(W)$. It supports some CE (x_1, y_1) with continuation promise $w_1(\cdot)$.
- W is self-generating, so $w_1(y_1)$ can be supported by some other CE (x_2, y_2) with promises $w_2(\cdot)$.
  - Importantly, every $w_1(\eta)$, $\eta \in Y$, can also be supported by some CE and continuation promises in W. This prescribes entire strategy $\sigma\big|_{(x_1, y_1)}$.

APS machinery

Theorem 1 (A self-generating set is a subset of V)

If $W \subset \mathbb{R}$ is bounded and self-generating, then $B(W) \subseteq V$.

- Iterating in this way, for each $v \in B(W)$ we construct sequences of $w_i(\cdot)$. Since $\delta < 1$ and W is bounded,

$$v = \lim_{t \to \infty} \left[(1-\delta) \sum_{s=1}^{t} \delta^{s-1} r(x_s, y_s) + \delta^t w_t(y_t) \right]$$
$$= (1-\delta) \sum_{s=1}^{\infty} \delta^{s-1} r(x_s, y_s).$$

- By construction $V_g(\sigma) = v$, so $v \in V$.

APS machinery

Proposition 2

The set V is compact.

Proof. V bounded $\implies$ $\mathrm{cl}(V)$ compact; $V \subseteq B(V) \subseteq B(\mathrm{cl}(V))$, so $\mathrm{cl}(V) \subseteq B(\mathrm{cl}(V))$. Theorem 1 implies $B(\mathrm{cl}(V)) \subseteq V$. $V = \mathrm{cl}(V)$ is compact.

Corollary (Two-value characterization)

A competitive equilibrium $(x, y) \in C$ is supported by an SPE, that is $x = \sigma_1^h$ and $y = \sigma_1^g$, if and only if there exist $(v_1, v_2) \in V \times V$ such that

$$(1-\delta)r(x,y) + \delta v_1 \geq (1-\delta)r(x,\eta) + \delta v_2, \qquad \forall \eta \in Y.$$

Proof.

- *Sufficiency*: Define $v(y) = v_1$ and $v(\eta) = v_2$ for $\eta \neq y$.
- *Necessity*: Set $v_1 = v(y)$; set $v_2 = \inf_{\eta \in Y} v(\eta) \in V$ since V is compact.

APS machinery

Collecting results, we know that

1. $V \subseteq B(V)$ (by Lemma 1).
2. If $W \subseteq B(W)$, then $B(W) \subseteq V$ (by theorem 1).

   $\Longrightarrow V = B(V)$.

3. B is monotone and maps compact sets into compact sets.

   $\Longrightarrow$ We can find V by iterating from W_0 such that $V \subseteq B(W_0) \subseteq W_0$.
   - To initialize W_0, set $W_0 = [\underline{w}_0, \overline{w}_0]$, where

$$\overline{w}_0 = \max_{(x,y)\in C} r(x,y), \qquad \underline{w}_0 = \min_{(x,y)\in C} r(x,y).$$

Implication: all SPE values can be recovered using recursive strategies.

Recursive strategies

Definition 11 (Recursive strategies)

A strategy is *recursive* if there is a tuple of functions $\phi = (z^h, z^g, \mathcal{V})$ and an initial condition $v_1 \in \mathbb{R}$ such that:

$$x_t = z^h(v_t), \qquad y_t = z^g(v_t), \qquad v_{t+1} = \mathcal{V}(v_t, x_t, y_t),$$

Definition 12 (Recursive SPE)

A recursive strategy (ϕ, v) is a *subgame perfect equilibrium* (SPE) if and only if $v \in V$ and

1. $(z^h(v), z^g(v))$ is a competitive equilibrium.
2. For each $\eta \in Y$, $\mathcal{V}(v, z^h(v), \eta) \in V$.
3. For each $\eta \in Y$,

$$\begin{aligned} v &= (1-\delta) r[z^h(v), z^g(v)] + \delta \mathcal{V}[v, z^h(v), z^g(v)] \\ &\geq (1-\delta) r[z^h(v), \eta] + \delta \mathcal{V}[v, z^h(v), \eta]. \end{aligned}$$

Taxation: a stick-and-carrot strategy

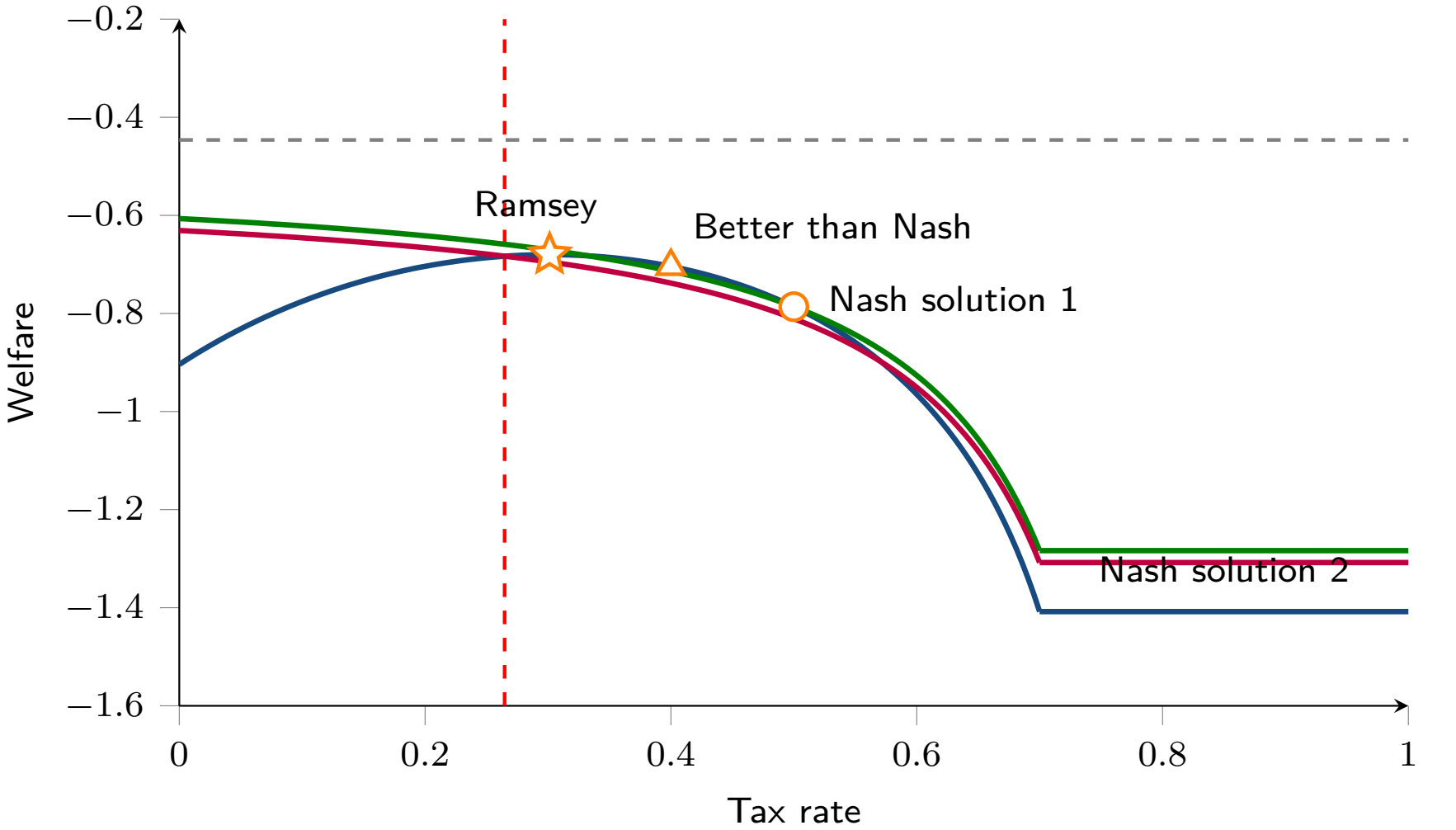
- A worse than Nash outcome can be sustained using a "carrot"
- Then, because this worse than Nash outcome can be used as a punishment, it can sustain something even better, possibly as good as Ramsey.
- Taxation example: $\alpha = 0.3$, $\varepsilon = 0.1$, $\delta = 0.20$.
  - Tax 0.6 today, Ramsey tomorrow if no deviation; otherwise restart. Let $\tilde{v} = (1-\delta)\, W^c(0.6) + \delta v^R$:

$$v^N \geq \underbrace{\tilde{v}}_{-0.9079} \geq (1-\delta) \underbrace{\max_{0 \leq \eta \leq 0.6} r[\ell(0.6), \eta]}_{-0.9613} + \delta \tilde{v} = -0.9506.$$

  - Using $\tilde{v}$ to punish deviations sustains Ramsey:

$$\underbrace{v^R}_{-0.6801} \geq (1-\delta) \underbrace{\max_{0 \leq \eta \leq 0.6} r[\ell(\tau^R), \eta]}_{-0.6418} + \delta \tilde{v} = -0.6951.$$

Taxation: stick-and-carrot punishment



Worst SPE values

Conclusion